LAB3: Modelling and Control of UAVs

Individual Deliverable 2

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1 A. Structure of Quadrotors

Let $c_f > 0$ and $c_d > 0$ be the lift and yaw-reaction coefficients, $d > 0$ the arm length, and $a = dc_f/\sqrt{2}$. Use an upward-z body frame and the motor numbering for which the functional vehicle has the following wrench map:

$$\begin{bmatrix} f \\ \tau_x \\ \tau_y \\ \tau_z \end{bmatrix} = F_a \mathbf{u}, \qquad F_a = \begin{bmatrix} c_f & c_f & c_f & c_f \\ a & a & -a & -a \\ -a & a & a & -a \\ c_d & -c_d & c_d & -c_d \end{bmatrix}. \tag{1}$$

Here $u_i = \omega_i^2 \geq 0$ describes the positive lift-producing operating direction of each mounted propeller. The alternating yaw signs already encode the different positive physical spinning directions.

For vehicle (a),

$$\det(F_a) = -16a^2 c_f c_d \neq 0, \qquad \operatorname{rank}(F_a) = 4. \tag{2}$$

Thus small changes of collective thrust and the three torques can be independently allocated around a feasible hover. Full rank alone does not guarantee feasibility for arbitrarily large commands under motor bounds.

In vehicle (b), propellers 3 and 4 are changed and their spin directions are reversed. They continue to produce upward lift, but their yaw-reaction signs reverse. With the same numbering and frame,

$$F_b = \begin{bmatrix} c_f & c_f & c_f & c_f \\ a & a & -a & -a \\ -a & a & a & -a \\ c_d & -c_d & -c_d & c_d \end{bmatrix}. \tag{3}$$

Its fourth row is $-c_d/a$ times its third row. The first three rows remain independent, so

$$\boxed{\operatorname{rank}(F_b) = 3, \qquad \tau_z = -\frac{c_d}{a}\tau_y = -\frac{\sqrt{2}c_d}{dc_f}\tau_y.} \tag{4}$$

Therefore yaw torque is coupled to pitch torque for this convention. A different documented motor order can change the row signs or the named horizontal torque, but not the rank deficiency.

For example, consider a level hover with zero angular velocity and a reference that holds position while requesting a yaw angular acceleration. Nonzero yaw torque necessarily produces the linked pitch torque, which tilts the thrust axis and disturbs horizontal position. Consequently vehicle (b) cannot generally track arbitrary independent references for $[x, y, z]$ and yaw. Some specially compatible motions can still be achieved.

2 B. Compensation of the Specified Drag

The empirical body-frame force in the exercise is quadratic in each velocity component and acts against that component. Let

$$D = \mathrm{diag}(0.1, 0.1, 0.2), \qquad \mathbf{v}^b = R^\top \mathbf{v}, \qquad \mathbf{F}^b_{\mathrm{drag}} = -D(\mathbf{v}^b \odot |\mathbf{v}^b|), \tag{5}$$

where R maps body vectors to world vectors, $\odot$ is the componentwise product, and the absolute value is componentwise. A linear expression $-D\mathbf{v}^b$ would describe a different force model.

The world-frame disturbance is

$$\mathbf{F}_{\mathrm{drag}} = R\mathbf{F}^b_{\mathrm{drag}}. \tag{6}$$

With upward z, the translational dynamics are

$$m\dot{\mathbf{v}} = -mg\mathbf{e}_3 + fR\mathbf{e}_3 + \mathbf{F}_{\mathrm{drag}}. \tag{7}$$

Define $\mathbf{e}_x = \mathbf{x} - \mathbf{x}_d$ and $\mathbf{e}_v = \mathbf{v} - \mathbf{v}_d$. If K_x, K_v are acceleration-feedback gains, the nominal commanded force is

$$\mathbf{F}_0 = m\left(\mathbf{a}_d + g\mathbf{e}_3 - K_x\mathbf{e}_x - K_v\mathbf{e}_v\right). \tag{8}$$

The model can be explicitly compensated by subtracting the estimated disturbance:

$$\mathbf{F}_{\mathrm{comp}} = \mathbf{F}_0 - \widehat{\mathbf{F}}_{\mathrm{drag}} \tag{9}$$

$$= \mathbf{F}_0 + RD(\mathbf{v}^b \odot |\mathbf{v}^b|) \quad \text{under the stated model.} \tag{10}$$

Use this compensated force consistently in both the desired thrust direction and scalar thrust projection:

$$\mathbf{b}_{3d} = \frac{\mathbf{F}_{\mathrm{comp}}}{\|\mathbf{F}_{\mathrm{comp}}\|}, \qquad f = \mathbf{F}^\top_{\mathrm{comp}} R\mathbf{e}_3. \tag{11}$$

The remaining desired body axes are constructed from $\mathbf{b}_{3d}$ and the desired yaw, as in the geometric controller.

As a sign check, take $R = I$ and $\mathbf{v} = [2, -1, 0]^\top$ m/s. Then

$$\mathbf{F}_{\mathrm{drag}} = [-0.4, 0.1, 0]^\top \text{ N}, \qquad -\mathbf{F}_{\mathrm{drag}} = [0.4, -0.1, 0]^\top \text{ N}. \tag{12}$$

The commanded correction opposes the disturbance. In the ideal force-tracking limit with an exact drag model, the extra dynamics term is cancelled, reducing the disturbance that feedback must reject. Attitude tracking, model errors, sampling delay, and actuator limits can still produce nonzero tracking error. This is a modelling proposal, not a claim of an experimentally measured before/after improvement.

Sources

MIT VNAV LAB3 exercise statement: `https://vnav.mit.edu/labs_2023/lab3/exercises.html`.

T. Lee, M. Leok, and N. H. McClamroch, "Geometric tracking control of a quadrotor UAV on SE(3)," *IEEE CDC*, 2010.